

ALCHEMUS

Advanced Drug Analysis Platform

Research Owner Wallet

ASD9BU3X3YDhVwiuB8eTuTmRLCyM45eZzL3vFr1wZT

Date: 5/1/2025

Compounds: Bismuth Subsalicylate, Caffeine, Chlorpheniramine, Dextromethorphan

RESEARCH PURPOSES ONLY

This analysis is generated for research and educational purposes. The predictions and insights provided should not be considered medical advice.

New Compound Analysis

Name: Bismuthcaffeamine dextroate

Formula: C₂₀H₃₆O₆N₂Bi

Weight: 600.51

Drug-likeness: 7.5

Activity:

Antidiarrheal, Antihistamine, Cough Suppressant, Stimulant

Target Analysis

H1 Histamine Receptor

Binding Affinity: -9.2

Involved in allergic reactions causing vasodilation and itching.

5-HT₃ Serotonin Receptor

Binding Affinity: -8.5

Plays a role in nausea and vomiting, potentially targeted by bismuth components.

Toxicity Profile

Risk Level: Moderate

Affected Systems: Liver, Kidneys, Central Nervous System

Mitigation Suggestions:

- Monitor liver enzymes
- Maintain hydration
- Consider lower doses in patients with CNS disorders

Pharmacokinetics

Absorption:

Moderate oral bioavailability

Distribution:

Widely distributed, crosses the blood-brain barrier

Metabolism:

Primarily hepatic via cytochrome P450 enzymes

Excretion:

Renal and fecal elimination

Half-life: 4-6 hours

Detailed Analysis

The newly synthesized compound Bismuthcaffeamine dextroate brings together the diverse therapeutic profiles of Bismuth Subsalicylate, Caffeine, Chlorpheniramine, and Dextromethorphan. Combining these components, it aims to provide antidiarrheal protection, antihistaminic allergy prevention, cough suppression, and mild CNS stimulation. Featuring a high molecular weight of 600.51 g/mol, the compound resembles a complex organic structure characterized by multiple aromatic rings and heteroatoms facilitating a wide variety of interactions at a molecular level. The binding affinity for H1 and 5-HT3 receptors underscores its potential to simultaneously curb allergic reactions and symptoms like nausea. In pharmacokinetic terms, the compound boasts moderate absorption, effective CNS penetration, and a multi-pathway clearance involving metabolic and renal routes. Despite its potential, moderate toxicity risks involving the liver, kidneys, and CNS, warrant further investigation, particularly concerning dose adjustments and specific patient population considerations. In the broader landscape of drug development, the prospect of reduced polypharmacy through this multifaceted agent stands out, albeit requiring meticulous preclinical and clinical evaluation to unlock its therapeutic promise. As with any complex polypharmacological entity, its development journey is accompanied by risks of metabolic stress, necessitating cross-disciplinary approaches for risk mitigation and optimization toward successful therapeutic application.

Combine base compounds, simulate new drug possibilities with AI, and earn rewards through Solana blockchain. Join our community of researchers and innovators in shaping the future of pharmaceutical discovery.

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